

AT-FSV / ATAI — CMFSB ISOLATION MODULE

EXECUTIVE TEASER — Path A Module License

Document 00 | Revision 5.0 | 2026-10-02 | Confidential — NDA controlled
Founder / Legal Owner: Amirbahador Atai

The problem in one line

Specialty, ambulance, defence-utility and luxury cabins still receive road vibration that degrades comfort, clinical stability and task performance. Soft mounts help ride but hurt handling; stiff mounts help handling but hurt ride.

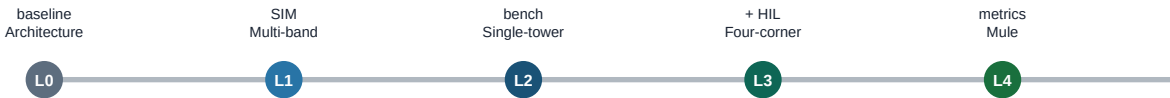
The solution concept in three lines

CMFSB keeps the upper body **mechanically captured at all times** — not free magnetic levitation. A macro stage (± 50 mm) isolates on a passive/fail-safe path; a micro stage provides high-bandwidth force in a controlled gap. Dual modes: **COMFORT** (soft isolation) and **SPORT** (rigid lock for handling) on the same module family.

What Path A is (and is not)

Path A is	Path A is not
Field-limited license / supply of isolation module ATFSV-MOD-001	A complete vehicle programme
Dual-mode SPORT/COMFORT architecture with defined fail-safe behaviour	A comfort-percentage guarantee without evidence
ICD-controlled interface for integration	Type approval or public-road certification
A diligence package with honest OPEN items and an evidence ladder	Granted patents or freedom-to-operate clearance

Evidence ladder — where we are today



Today: L0 complete. Comfort claims require L1+. Term sheet strength rises at L2.

What we explicitly do not claim in this document

No measured comfort percentage without configuration-managed tests. No certification. No production cost or royalty figures in this technical baseline. No free levitation. No patent grant. TRL 2–3 — preliminary engineering baseline only.

The full technical baseline follows on the next pages (R4.9 body retained in full, with R5.0 executive and wow-factor front matter added — nothing removed).

W1. Evidence Snapshot — What Is Closed vs What Buyer Funding Unlocks

Claim / capability	Status today	Next proof	Gate
Mechanical capture invariant	BASELINE written	Fixture / HIL	G2
Power-loss → passive support (S4)	BASELINE policy written	Single-tower L2 demo	G3
Dual-mode SPORT lock confirm	TARGET ≤ 3 s	Lock cycle test	G3
Ride isolation (COMFORT)	OPEN	Multi-band SIM L1	G2
COMFORT handling limits	Policy required (numeric)	Dynamics freeze + sim	G2
Module mass ≤ 55 kg	TARGET	CAD + weigh L2/L3	G3
Force authority ± 2.5 kN	STUDY	Force-gap curve L2	G3
Stop energy path	Numeric REQ written	STR-001 curve	G3
Cabin field plan ≤ 0.5 mT	TARGET plan	Map at G5	G5

How to read this table: BASELINE means architecturally committed and written; TARGET needs verification; OPEN needs an owner and a funded product. Comfort marketing language is inappropriate before L1.

W2. How CMFSB Compares to Familiar Options (qualitative — no brand names)

Attribute	Soft cabin mounts	Active cabin / seat	CMFSB module
Large body travel (± 50 mm class)	Limited	Usually limited	Yes — macro stage
Fail-safe if power lost	Passive remains	Often degrades	S4 passive + prefer lock
Dual-mode handling lock	No	Rare / not primary	Yes — SPORT lock
Magnetic as sole vertical path	No	No	Forbidden by invariant
ICD module boundary (Path A)	Usually vehicle-tied	Often seat/cabin tied	Yes — licensable module
Buyer can evaluate without full vehicle	Sometimes	Sometimes	Yes — Path A intent

This table is qualitative orientation for executives. It is not a ranked competitive score and does not assert measured superiority.

W3. Twelve-Month Decision Path (Path A — contingent on funding)



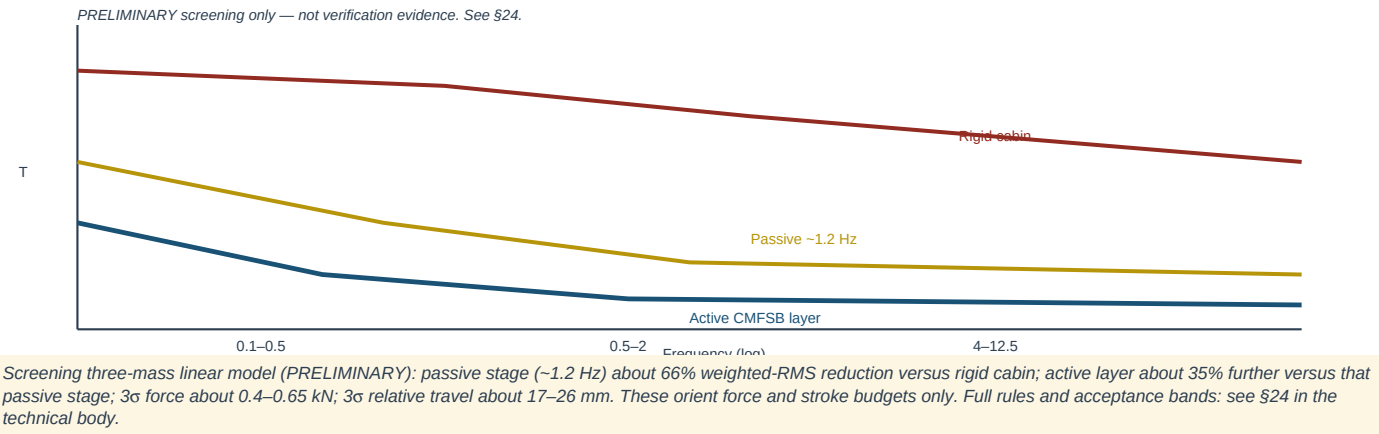
NDA and Doc 00 review can start immediately (T1). Meaningful term-sheet discussion is typically after L1 multi-band SIM and preferably after L2 single-tower force–gap and power-loss demonstration. Dates assume committed budget; without funding the schedule is not valid.

W4. What You Buy on Day 1 vs What You Still Fund

With a Path A license / supply agreement you obtain	You still fund or co-fund (typical)
Rights to the module architecture within the contracted field	L1 multi-band SIM against agreed passive reference
Dual-mode SPORT/COMFORT design intent and state-machine rules	L2 single-tower bench (force, power-loss, lock)
ICD-controlled mechanical/electrical/safety/lock interfaces	L3 four-corner / HIL if vehicle integration is in scope
Written S4 fail-safe behaviour and capture invariants	Application-specific calibration and vehicle integration
Honest OPEN/TARGET register and evidence ladder	Regulatory path for the end vehicle (Path B / buyer)
Diligence package (this Document 00 under NDA)	Production tooling, costed BOM, quality system

Transparent split of ‘what is designed’ versus ‘what is still evidenced’ is intentional. It protects both sides from comfort or schedule over-claim.

W5. PRELIMINARY Isolation Picture (screening — not proof)



W6. Top Risks → Mitigation → Buyer Action

Risk	If left open	Our mitigation	Buyer action that helps
R-01 Capture loss	No safety credit	Invariants; FMEA; REQ-M01/14	Require L2/L3 demos in term sheet
R-02 Comfort not shown	No % claims	SIM-001 multi-band rules	Co-fund L1; freeze passive k,c
R-03C COMFORT handling	Soft in manoeuvre	Numeric %g + anti-roll policy	Agree dynamics limits early
R-04S S4 at speed	Soft handling after power loss	Lock preference; limp-home	Review HARA row at G2
R-08 Stop energy	Structural over-stress	REQ-M09; STR-001	Require curve before G3

W7. IP Posture (non-legal summary — not advice)

Differentiation emphasised for counsel discussion: mechanical capture invariants, macro–micro split with defined stiffness constants, dual-mode lock with positive confirmation, and module ICD boundary — not predictive mode-select UX alone (crowded prior art). Provisional filings and prior-art search status are maintained outside this document in counsel-controlled files (T3). This Document 00 does not assert patentability, claim scope, or freedom-to-operate. Chain-of-title and assignment logs are programme-controlled outside Doc 00.

Recommendation before broad enabling disclosure beyond T1: complete counsel prior-art screen on lockable cabin / predictive suspension themes (see §17).

W8. How Executives Should Read the Rest of This Package

Pages after this executive front matter contain the full R4.9 technical baseline retained without deletion: requirements, state machine, FMEA, risks, dual-mode detail, SIM skeleton, mass/energy appendix, and corrections index. Engineers should start at §0 vocabulary and §2 invariants. Commercial staff should stay with Path A language and ignore Appendix A unless Path B is in scope.

Reader	Start here	Then
CEO / BD	This teaser + W1–W4	§13 paths; §23 commercial framework
CTO / Chief Engineer	§2 invariants; §8 S4; §15 risks	§7 REQ; §24 SIM
Safety	§8–§11; FMEA RPN	HARA; limp-home policy
Procurement / legal	W4 buy vs fund; §16 tiers	NDA; ICD; T2/T3 boundary

W9. Presentation and Control Notes for R5.0

R5.0 = R4.9 technical body preserved in full (no text or table removed) + this executive / wow-factor front matter (sections W1–W9). PDF metadata title set to ATFSV-DOC-00-R5.0. Footer carries revision and page number on every page. Distribution remains controlled under NDA for T1. Empty pages are not used as separators.

— End of R5.0 Executive Front Matter — Full technical baseline (R4.9 body) continues below without omission —

AT-FSV / ATAI — CMFSB ISOLATION MODULE

ATFSV-MOD-001 | Captured Mechanical Floating Safety Body

DOCUMENT 00 — MASTER ENGINEERING BASELINE

Revision 4.9 | Module-Primary External Baseline | Dual-Mode Comfort / Sport | 2026-10-02

Founder / Legal Owner: Amirbahador Atai

Module Datasheet Summary (ATFSV-MOD-001)

Parameter	Value	Status
Architecture	Mechanically captured upper body + macro–micro isolation (NOT free levitation)	BASELINE
Dual operating modes	SPORT (rigid lock / handling) + COMFORT (soft isolation) — see §22	BASELINE
Passenger cell / roll (Path B)	Summary only in Doc 00; full carbon-cell / roll / 4×4–6×6 detail in ATFSV-DOC-01	OUT OF BASELINE detail
Mode selection paths	Driver command / radar-based / GPS–map-based (three-path arbitration)	TARGET
Reference topology	Hybrid hydraulic (macro) + Lorentz (micro)	BASELINE path
PM+EM reluctance	OPTION only — stability gate; gap-dependent bias at min gap 15 mm (see §4)	OPTION
Static load classes	Size S/M/L with CORRECTED 3 g bands in §5.2	STUDY
Macro stroke	±50 mm body-relative (100 mm total range)	TARGET
Micro travel (Lorentz)	±7.5 mm stroke; coil–magnet gap few mm (ICD)	TARGET
Peak dynamic force	±2.5 kN per tower (study); higher after proof	STUDY
Dynamic force (RMS continuous)	≈0.7 kN RMS planning — may be oversized vs road-noise need (see §0.2)	STUDY
Module mass boundary	Core tower ≤55 kg TARGET; full corner STUDY	TARGET / STUDY
Body lock (SPORT)	Mechanical / hydraulic rigid engagement; upper body seats on lower body	TARGET
Primary commercial path	Path A — module field license / supply	BASELINE
Baseline operational GVW (ref. vehicle)	7,500 kg single baseline (Appendix A)	TARGET

R4.9 is the R4.8 module-primary baseline with incorrect numerical values corrected in place, drawn figures added, missing numeric/traceability content completed, Path B detail moved to summary, and commercial sensitivity reduced. Structured tables and engineering explanations from R4.8 are retained. TARGET, STUDY, OPEN and ASSUMPTION are not compliance claims.

0. Document Control and Status Vocabulary

Field	Value
Document ID / Revision / Date	ATFSV-DOC-00-R4.9 / 4.9 / 2026-10-02
Supersedes	ATFSV-DOC-00-R4.8
Legal Owner	Amirbahador Atai
Document Owner	ATFSV Project Engineering Documentation Authority
Author / Reviewer / Approver	ATFSV Engineering / TBD / TBD
Distribution	Controlled — NDA / due-diligence / supplier engagement
Classification	Confidential — IP / Engineering Development
Configuration status	NOT configuration-controlled until a CM system is established
Purpose	Module architecture baseline, dual-mode Comfort/Sport, mass/energy analysis, safety planning, supplier engagement, NDA due diligence
PDF metadata title	ATFSV-DOC-00-R4.9 CMFSB Module-Primary External Baseline

Chain of title and disclosure logs are maintained outside this document. External release follows the tiered scheme in §16. This document is not legal advice.

0.1 Status Vocabulary (exclusive)

Label	Mandatory meaning
CONTROLLED DESIGN LIMIT	Architecture ceiling; not a measured result.
BASELINE	Included in current architecture accounting.
TARGET	Numerical target requiring verification.
STUDY	Study allocation or scenario; not frozen design.
ASSUMPTION	Modelling assumption needing confirmation.
DERIVED	Calculated from stated assumptions.
OPTION	Not credited in baseline safety or mass closure.
OPEN	Unresolved; owner and gate required.
TBD	Not responsibly fixed at this revision.
PRELIMINARY	Indicative analysis — not verification evidence.
REFERENCE	Context only; not a Path A prerequisite.
OUT OF BASELINE	Maintained in another controlled document (e.g. ATFSV-DOC-01).
PROPOSED	Application or commercial proposal — not a design freeze.
VERIFIED	Reserved for objective CM evidence only — not used in R4.9.

Rule: TARGET, STUDY, OPEN and ASSUMPTION are not compliance claims. A value becomes a controlled design requirement only after requirement ID, owner, verification method and evidence reference are baselined.

Naming governance (Finding 15): Phase = Ph0–Ph3; Gate = G0–G5; Priority = Pri-1 to Pri-3; Evidence ladder = L0–L4. Do not use one token (P0/P1/P2) simultaneously for phase, gate and priority.

0.2 Value Statement (screening)

Problem. Occupant compartments in specialty, defence-utility, ambulance and luxury extreme vehicles receive road and terrain vibration that degrades comfort, clinical stability, task performance and long-duration fatigue. Conventional suspensions couple road input into the body across ISO 2631-1 weighted bands and the motion-sickness band (approximately 0.1–0.5 Hz). Soft isolation alone can degrade handling in rapid manoeuvres (e.g. moose test).

Solution concept. CMFSB keeps the upper body mechanically captured at all times. Macro travel (±50 mm body-relative, 100 mm total range) provides large-stroke isolation on a passive support path with active leveling. Micro travel (±7.5 mm) provides high-bandwidth force control via the Lorentz stage. Static load is never credited to magnetic force. Dual operating modes — SPORT (upper body seats and locks to lower body for handling) and COMFORT (upper body isolates for soft ride) — with three selection paths (driver, radar, GPS/map). “Floating” means body isolation relative to the chassis — not free levitation and not marine flotation.

Criterion	R4.9 statement	Status
Ride isolation (COMFORT)	Multi-band transmissibility vs defined passive hydro-pneumatic reference in 0.1–0.5 / 0.5–2 / 4–12.5 Hz	OPEN — ATFSV-SIM-001
Handling (SPORT)	Upper body locked/seated; effective roll/yaw stiffness approaches rigid-body class	TARGET — §22
Mode transition time	≤3 s typical TARGET (full SPORT↔COMFORT under defined conditions)	TARGET
Module mass (core tower)	≤55 kg TARGET (structure + Lorentz + sensors + local electronics)	TARGET
Dynamic force (peak)	±2.5 kN study per module	STUDY
Dynamic force (RMS continuous)	≈0.7 kN RMS planning — may exceed road-noise need; leveling/manoeuvre sized separately	STUDY
Static support	Passive/fail-safe path ≥5.5 kN class study per module	STUDY
Power-loss behaviour	Defaults to passive support within design envelope (S4)	BASELINE invariant

Correction (Finding 2) — multi-band SIM and screening results: A single $\geq 10\%$ floor versus an undefined passive reference is not sufficient. ATFSV-SIM-001 shall declare passive body-support k and c (about 1.0–1.4 Hz class — not magnet-carriage stiffness) and at least ISO 8608 class B and class C roads at stated speeds. Acceptance is reported in bands 0.1–0.5 Hz (motion sickness), 0.5–2 Hz (bounce) and 4–12.5 Hz (ISO 2631-1 weighted content), each with a minimum TARGET frozen in the simulation plan. Screening three-mass linear model (PRELIMINARY — not verification evidence): passive stage (~1.2 Hz) about 66% weighted-RMS reduction versus rigid cabin; active layer about 35% further versus that passive stage; 3σ force demand about 0.4–0.65 kN; 3σ relative travel about 17–26 mm. The $\geq 10\%$ floor is therefore conservative. A 0.7 kN RMS force budget is likely larger than road-noise need; leveling and manoeuvre loads are sized separately.

1. Engineering Baseline and Controlled Values

R4.9 defines a mechanically captured isolation architecture. The upper occupant/safety body remains structurally constrained at all times. Static load is carried by compliant mechanical and hydraulic support. Magnetic or Lorentz force is a bounded dynamic-control layer. Dual modes are COMFORT (soft isolation) and SPORT (rigid lock). Vehicle-level mass, powertrain and range data live in Appendix A as REFERENCE only and are not prerequisites for Path A module evaluation.

Parameter	R4.9 value	Status
Upper-body reference mass	2,150–2,350 kg study band	STUDY
Reference topology	Hybrid hydraulic macro + Lorentz micro	BASELINE path
PM+EM	OPTION with stability gate; gap-dependent bias	OPTION
Macro travel	± 50 mm body-relative	TARGET
Micro travel / gap	± 7.5 mm / 15–30 mm	TARGET
Peak dynamic force	± 2.5 kN per tower study	STUDY
SPORT lock	Rigid engagement; ≤ 3 s TARGET; positive confirmation	TARGET
Baseline operational GVW (ref. vehicle)	7,500 kg single baseline	TARGET

2. System Architecture and Safety Invariants

The architecture separates chassis, macro isolation stage, micro actuation carrier and upper body. The separation is functional and structural. It is not a claim of free levitation and not a claim of marine flotation. Progressive stops and capture geometry remain continuous through the stroke. These invariants are the primary reason the architecture can be discussed with safety-minded buyers without implying an unconstrained floating cabin.

Safety invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture.
Static support	Vehicle weight is supported by passive or fail-safe compliant load paths.
Magnetic control	Magnetic force is bounded and cannot be the sole vertical load path.
Loss of electrical power	System transitions to a mechanically stable configuration without requiring live software.
Crash load path	Progressive stops and structure carry crash loads without a closed magnetic loop.
Locking	SPORT lock is not the primary crash arrestor; stops remain the ultimate load path.
Free-float ban	Unconstrained free-floating upper body is forbidden in all transitions.

Figure 1 — Layer stack and primary load path (Finding 1)

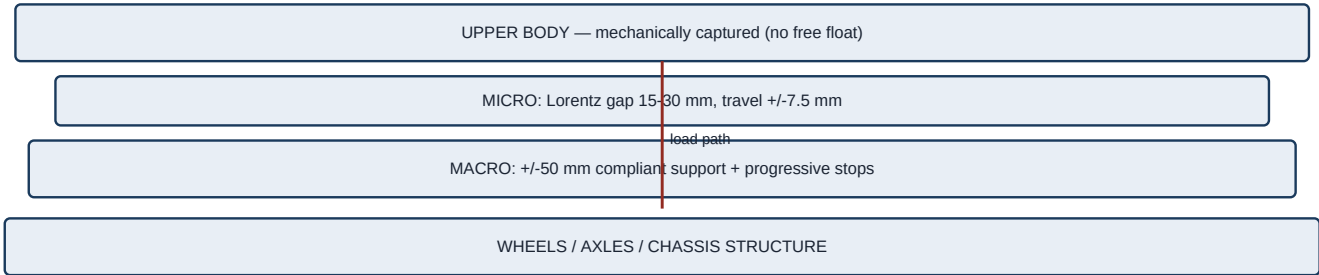


Figure 1. Drawn schematic. CAD three-views remain a programme deliverable (FIG-CAD-01).

3. Macro–Micro Travel, Stiffness Stages and Leveling

A direct-mount magnet pair cannot provide ± 50 mm body travel and a 15–30 mm magnetic gap at the same time. The macro stage provides large body-relative travel on a compliant support. The micro stage rides on a local carriage that maintains the working gap for the Lorentz (or OPTION PM+EM) actuator. Body-support stiffness for ride shall be sized near 1.0–1.4 Hz for the upper-body mass share; local magnet-carriage stiffness may be substantially higher and must not be confused with ride spring rate.

3.1 Frequency, deflection and leveling rationale

At upper-body mass share $\approx 2,250$ kg (≈ 562 kg per corner), a body-support stiffness of ≈ 40 kN/m yields natural frequency ≈ 1.34 Hz. Static deflection under 1 g is ≈ 25 mm. A payload change of 100 kg per corner (≈ 0.98 kN) at $k \approx 40$ kN/m produces ≈ 25 mm additional deflection; without leveling, stroke margin erodes. Four-corner leveling settle ≤ 3 s TARGET class is therefore required to restore ride height after load change. Carriage local stiffness (e.g. 133–200 kN/m class) is for gap stability and must not be used as the body bounce spring in SIM-001.

Stage	Role	Envelope / sizing	Status
Macro body support	Primary ride isolation	Study ~1.0–1.4 Hz (not magnet-carriage k)	OPEN sim
Local magnet carriage	Gap / actuator mounting	May be stiff locally (e.g. ≥ 133 –200 kN/m class)	STUDY
Micro control	Gap regulation / force	± 7.5 mm; bandwidth ≥ 20 Hz TARGET	TARGET
Progressive stop	Energy limit after excursion	Force–displacement from design/test	OPEN
Leveling	Height set and load change	Rate-limited; four-corner settle ≤ 3 s TARGET class	TARGET

3.2 Macro–Micro Force Path

Static and low-frequency load travels through the hydraulic or passive macro path. High-bandwidth correction is applied by the Lorentz micro path within the gap envelope. Progressive stops engage on abnormal excursion. Capture geometry remains continuous. Optional rate-limited hold is not the primary crash path and must not be credited as a substitute for stops.

Figure 2 — Force path (conceptual)

Chassis → macro compliant support / leveling → micro carriage (gap maintained) → upper body. Stops parallel to macro path. Lorentz force is bidirectional within ± 2.5 kN study and does not carry static weight.

4. Actuator Topology, PM Bias Rule and Power Budgets

The reference development path is hybrid hydraulic macro support plus Lorentz micro force. PM+EM reluctance actuation remains an OPTION that is not credited in baseline safety or mass closure until a stability gate is passed with measured force–gap data.

Topology	Role	R4.9 position
Hybrid hydraulic + Lorentz	Macro support + micro force	REFERENCE PATH (BASELINE)
PM + EM reluctance	Bias + current-controlled force	OPTION — stability gate required

4.1 Lorentz continuous force and power (screening)

Lorentz continuous force and copper-loss screening numbers remain STUDY pending bench data. The previous line “≤0.5 kW” described copper loss only and is incomplete as a total module power budget. Controllers, sensors, hydraulic pump average power, leveling flow and lock pulses must be included before L2 closure.

Correction (Finding 12) — gap-dependent PM bias for OPTION path: Negative stiffness increases as gap falls. Evaluate at minimum operating gap 15 mm using $F_{PM}(15\text{ mm}) \leq k_{gap} \times 0.015 / 3$ for a margin factor of 1.5 on negative stiffness. Example: $k_{gap} = 200\text{ kN/m} \rightarrow F_{PM} \leq 1.0\text{ kN}$ at 15 mm. Equivalent force at 22.5 mm depends on the force–gap law (illustrative order ~0.44 kN under inverse-square scaling — confirm with measured curve). A bare CONTROLLED DESIGN LIMIT of ≤1.5 kN without stating gap is incomplete and is superseded for design use.

Correction (Finding 10) — full module power and hydraulic budget:

Consumer	Budget note	Status
Lorentz copper loss (I²R)	Part of prior ≤0.5 kW discussion — not total module power	STUDY
Hydraulic pump average	Leakage, leveling duty cycle, ambient temperature	OPEN
Four-corner settle ≤3 s	Flow from piston area × stroke / time; sizes pump and accumulator	OPEN calculation
Sensors and controllers	12 V / 48 V electronics continuous draw	OPEN
Lock actuation pulse	Energy per engage — not a continuous load	OPEN
Piston area at 5.5 kN, 150–250 bar	Approximately 221–368 mm²	DERIVED screening
Accumulator	Support leak-down and settle pulses	OPEN sizing
48 V peak current	Must be justified from pulsed loads; do not assume continuous 131 A	OPEN

5. Mass Architecture, Corner Breakdown and Structural Loads

5.1 Per-corner and module boundary

The Path A commercial object is the module. Core tower mass is TARGET ≤55 kg including local sensors unless the ICD moves sensors to the body side. Full corner mass including unsprung share is STUDY and is closed in the vehicle mass book (ATFSV-MASS-001).

Element	Value	Status
Core module (tower)	≤55 kg TARGET	TARGET
Local sensors / harness share	Included in TARGET unless ICD shifts boundary	STUDY
Full corner with unsprung share	See rebuilt vehicle lines in Appendix A	STUDY

5.2 Module structural load cases — CORRECTED 3 g values

Correction (Finding 9): Previous 3 g bands 12–15 / 18–21 / 22–24 kN were inconsistent with the stated mass shares. Corrected bands follow. Static support path STUDY ≥5.5 kN remains the module datasheet floor for Path A sizing discussions.

Class	1 g support class	3 g load (corrected)	Prior value (superseded)	Status
S	3–5 kN class	9–15 kN	12–15 kN	STUDY
M	5–7 kN class	15–21 kN	18–21 kN	STUDY
L	7–8 kN class	21–24 kN	22–24 kN	STUDY

6. Tower Module Architecture (ATFSV-MOD-001) — Path A Primary Product

The module is the primary licensable unit under Path A. Buyers can evaluate Path A without funding a full 7.5 t vehicle programme. Functional envelope includes static support, dynamic force, stroke, gap, mass, bandwidth, life, power-loss behaviour, contamination protection and the SPORT lock interface. ICD domains: mechanical, hydraulic, electrical, communications, safety, lock.

Characteristic	Target / limit	Status
Static support path	≥5.5 kN class STUDY	STUDY
Dynamic force	±2.5 kN STUDY	STUDY
Bandwidth	≥20 Hz TARGET	TARGET
Mass	≤55 kg TARGET	TARGET
Life	≥10,000 h TARGET duty as defined in ICD	TARGET
Power loss	Passive support remains; high force inhibited	BASELINE
Contamination protection	Bellows/seals; detection; environmental test	TARGET
ICD domains	Mechanical, electrical, communications, safety, lock	OPEN draft

7. Module Requirements (owned) — ATFSV-MOD-001

R4.8 listed REQ-M01...M20 with qualitative freeze language. R4.9 retains the requirement set and ADDS numeric content where available, plus risk link, ICD domain and gate (Finding 13). Additional requirements M21–M23 close COMFORT handling, power budget and contamination gaps.

ID	Requirement	Risk	ICD	Gate
REQ-M01	Geometric capture in all baseline states	R-01	Mech	G1
REQ-M02	Passive static path; magnetic not sole support	R-01	Mech	G1
REQ-M03	Reference topology = hydraulic + Lorentz until change control	R-04	Mod	G1
REQ-M04	Macro ±50 mm; micro ±7.5 mm; gap 15–30 mm	R-02	Mech	G3
REQ-M05	Body-support stiffness for ~1.0–1.4 Hz study	R-02	Sim	G2
REQ-M06	±2.5 kN study; ≥20 Hz; ≤55 kg; ≥10,000 h	R-05	Mod	G3
REQ-M07	Static support ≥5.5 kN study path	R-05	Mod	G3
REQ-M08	Multi-band SIM-001 acceptance per §24	R-02	Sim	G2
REQ-M09	Stop stroke ≥50 mm beyond free play or justified energy; force ceiling ~441 kN total for 20 g × 2,250 kg class; S5 reduces free play	R-08	Struct	G3
REQ-M10	Power-loss → S4; lock preference and limp-home per §8	R-04S	Safety	G2
REQ-M11	Thermal within ICD limits	R-05	Mod	G3
REQ-M12	Cabin B-field plan; ≤0.5 mT planning target at seats pending guidance confirm	R-10	EMC	G5
REQ-M13	ICD mech/elec/comm/safety/lock released	R-13	Sys	G1
REQ-M14	No single failure causes loss of capture	R-01	Safety	G2
REQ-M15	Manual recovery / rescue access provisions	R-01	HF	G4
REQ-M16	Hold rate-limited; not primary crash path	R-08	Safety	G2
REQ-M17	SPORT lock engages with positive confirmation; ≤3 s	R-03	Lock	G3
REQ-M18	COMFORT mode provides full macro envelope without lock	R-03C	Mech	G3
REQ-M19	Mode arbitration priority table frozen	R-12	Ctrl	G3
REQ-M20	Transition never enters unconstrained free-float	R-01	Safety	G2
REQ-M21	COMFORT roll gradient / lateral-g policy numeric	R-03C	Dyn	G2
REQ-M22	Module electrical + hydraulic power budget closed at L2	R-14	Mod	G3
REQ-M23	Gap/lock seals + contamination detection	R-11	Mod	G3

8. Safety State Machine S0–S8 — States and Transitions

The state machine is the safety spine of the module. Thresholds remain PRELIMINARY until HIL and fixture tests freeze them. No transition may enter unconstrained free-float. Module supply is 48 V class.

Figure 3 — State overview



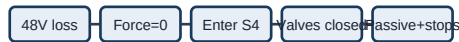
Figure 3. Drawn state overview (Finding 1).

State	Function	Notes
S0	Boot / self-test	Inhibit high force
S1	Standby / capture confirmed	Ready for mode enable
S2	Normal isolation (COMFORT)	6-DOF control; capture mandatory
S2S	SPORT locked	Lock confirmed; handling stiffness rises
S3	Soft degrade	Reduced authority
S4	Power-loss capture	Passive support; valves fail-closed
S5	Impact preparation	Prepare stops; reduce free play
S6	Impact / arrest	Progressive stop energy absorption
S7	Post-event safe	Isolation; rescue access
S8	Maintenance lockout	Physical lockout device

8.1 Transition table (PRELIMINARY thresholds)

Transition	Condition	R4.9 note
S0→S1	Self-test pass	—
S1→S2	Capture OK; operator enable; default COMFORT	COMFORT policy §22
S2↔S2S	Lock engage/release with confirmation; ≤3 s TARGET	Sensor confirm required
S2/S3/S2S→S4	Power loss detected; valves fail-closed	Hardware-dominant
S2→S5	Impact prediction / severe event	Reduce free play (Finding 5)
S5→S6	Stop engagement	Event-driven
S6→S7	Motion settled	No auto re-enable to S2
S4→S1	Power restored; sensors healthy	Self-test required
S4→S2	Forbidden direct	Only via S1
Any→S8	Physical lockout	Manual only

Figure 4 — Power-loss path (BASELINE invariant)



On loss of module 48 V supply or intentional isolation: active Lorentz force goes to zero; the system enters S4; the macro stage settles on passive hydraulic or pneumatic support within the design envelope; leveling valves fail-closed so height is retained; progressive stops limit residual motion. Software aliveness is not required for this path. Return to S2 only via S1 after self-test and explicit operator enable.

Correction (Finding 4) — S4 behaviour at speed: After power loss the body may remain on a soft passive support while vehicle speed is still high, which reduces handling stiffness. BASELINE preference: command lock engagement at S4 entry when residual energy allows. If lock cannot be confirmed, remain on passive support and apply limp-home speed limit TARGET ≤80 km/h class until a safe state is confirmed. A HARA/FMEA row for “loss of handling stiffness at speed after power loss” is mandatory. Prefer fail-toward-locked when actuator energy permits; document residual risk if unlock is stuck.

9. Summary FMEA (expanded)

R4.8 listed failure modes with severity, occurrence and detection columns but did not publish RPN. R4.9 adds RPN and strengthens contamination mitigations. Lorentz hardware that includes permanent magnets can attract ferrous debris; seals, bellows, detection and environmental testing are Path A requirements.

Failure mode	S	O	D	RPN	Detection / mitigation
Carriage jam	9	3	4	108	Gap/stroke sensor; stops; passive support
Link fracture	9	2	5	90	Capture frame; stops; redundancy analysis
Carriage drive loss	7	3	4	84	Fallback passive; S3
Leveling stuck / delayed	8	3	3	72	Level sensor; S3
Hose leak / burst	9	3	3	81	Pressure sensor; progressive stop
Stop wear	7	3	5	105	Inspection interval; design margin
Gap contamination (mud/ice/ferrous)	7	4	4	112	Bellows/seals; contamination detect; environmental test
Coil overheating	7	3	3	63	Temp sensor; thermal model
Lock contamination / no engage	8	3	4	96	Confirm sensor; inhibit SPORT claim
SPORT lock fails to engage	6	3	2	36	Lock status sensors; driver alert; stay S2
SPORT lock fails to release	5	3	2	30	Release sensor; limp SPORT; service
Partial lock (1–2 of 4)	8	2	2	32	Per-lock status; inhibit S2S if incomplete
False SPORT from radar/GPS	4	3	3	36	Driver override; confidence thresholds
False COMFORT in high lateral demand	7	2	3	42	Lateral-g / yaw rate inhibit; driver SPORT
OTA unsafe change	8	3	4	96	Signed updates; safety partition

Severity/Occurrence/Detection are screening 1–10 labels. Full HARA with ASIL candidates and FTA for uncontrolled separation and for partial-lock asymmetry are required. Leveling valves shall fail-closed on power loss. Magnet demagnetisation on the Lorentz path reduces Km (force constant), not a bias force.

10. Preliminary HARA (screening — PRELIMINARY)

Screening HARA remains PRELIMINARY and does not replace a full ISO 26262 work product. R4.9 adds a mandatory item: loss of handling stiffness at speed after transition to S4, linked to the §8 lock and limp-home decision. Controllability and driver warning concepts for that case shall be recorded before G2.

Hazard	Scenario	S/E/C (cand.)	Safety goal (draft)
Loss of geometric capture	Carriage/link failure at speed	S3/E3/C3	Maintain capture under single failure
Uncontrolled vertical motion	Force command runaway	S2/E3/C2	Bound force; enter S3/S4
Loss of static support	Hydraulic burst + stop failure	S3/E2/C3	Progressive stops carry residual load
Loss of handling stiffness at speed	S4 entry while COMFORT at highway speed	S2–S3 / E3 / C2–C3	Lock preference or limp-home; driver indication

11. Crash and Emergency Load Paths (screening)

Crash loads are carried by progressive stops and structure, not by closed-loop magnetic force. Optional rate-limited hold is not the primary crash path.

Correction (Finding 5) — numeric stop requirements: With ±50 mm free play and a severe pulse, relative contact speed can reach about 4.4 m/s; energy about 22 kJ for a 2,250 kg upper-body reference is a screening order of magnitude. REQ-M09a: progressive stop stroke capability ≥50 mm beyond nominal free play, or justified equivalent energy capacity. REQ-M09b: peak stop force limited to protect structure — screening ceiling about 441 kN total for 20 g × 2,250 kg, distributed by design. REQ-M09c: on S5 entry, free play shall be reduced by lock and/or hydraulic close when a crash precursor is confirmed. ATFSV-STR-001 shall produce the energy/stroke curve before G3.

12. Functional Safety, Cyber, Environment, Access and Service

Access and service provisions must allow rescue and maintenance without defeating capture in uncontrolled ways. Environmental sealing is mandatory for gap and lock interfaces (see FMEA RPN 112). Cybersecurity and software-update controls apply if on-board software can change force limits, mode logic or lock behaviour.

12.1 Standards and references (planning list)

Instrument	Use	Status
ISO 26262	Functional safety process candidate	Candidate
ISO 21448 (SOTIF)	Safety of intended functionality	Candidate
ISO 2631-1	Whole-body vibration evaluation	Candidate
ISO 8608	Road surface profiles for simulation	Candidate
ISO/SAE 21434	Cybersecurity engineering	Candidate
UN R13 / R79	Braking / steering if vehicle Path B	Candidate
UN R155 / R156 / R10	Cyber / software update / EMC	Candidate
Hydraulic machinery safety rules	Reference path fluid systems	Market-specific
Vehicle category + speed limiter	Mass class may invoke limiter regimes — REG-001	OPEN
Cabin magnetic field	Planning target ≤0.5 mT at seating positions pending formal guidance confirmation	TARGET plan
Ambulance (if App. A)	EN 1789; EN 1865; NFPA 1917 / CAAS GVS (applicability confirmed by buyer)	Buyer-specific

13. Dual Commercial Paths, Timeline and Reference Application

Path A (primary): Isolation Module ATFSV-MOD-001 — field license or supply, including dual-mode SPORT/COMFORT capability. Path B: Full vehicle partnership (detail in ATFSV-DOC-01; summary in §25). Path A does not require Appendix A powertrain or GVW figures.

13.1 Unified timeline (Path A) — contingent on funding

Phase	Month	Deliverable	Acceptance criterion
Ph0 — Kinematics & ICD freeze	0–2	3-D envelope, hard-stops, leveling, lock faces, ICD draft	Full stroke without clash; lock clear; ICD review
Ph1 — Single-tower bench	2–5	Force, thermal, power-loss, leveling, lock cycle	Force curve; passive landing; RMS thermal; lock engage/release
Ph2 — Four-tower rig	5–9	Four towers + representative mass + mode transitions	6-DOF; SPORT↔COMFORT ≤3 s; power-loss; stop impact
Ph3 — App. A mule (optional)	9–15	Module on donor chassis	ISO 2631-1 + handling comparison; structured evaluation

NDA evaluation after Ph0. Meaningful Path A negotiation after Ph1. Full module licence discussions typically require Ph2. Dates assume committed budget; without funding the schedule is not valid. Target funded Ph0 to first serious Path A discussion: 6–12 months.

14. Evidence Ladder (planning)

Level	Product	Gate link	Notes
L0	Architecture and this baseline	G0–G1	Consistency checks
L1	ATFSV-SIM-001 multi-band versus passive reference	G2	Required before comfort marketing
L2	Single-tower force-gap and power-loss demo	G3	Preferred before module term sheet
L3	Four-corner fixture + state-machine HIL	G4	Before vehicle integration commitments
L4	Instrumented mule versus passive baseline	G5	Before performance warranty language

15. Key Technical Risks (owned)

Correction (Finding 14): The risk register is expanded. R-03 is lock failure. PM force-gap instability applies only to the OPTION path and must not be attributed to Lorentz. Missing risks for COMFORT handling, S4, stop energy, prior art, title, cost, supply, contamination, electrical peaks, regulation and export control are added.

ID	Risk	Sev.	Mitigation	Gate
R-01	Loss of capture	Cat.	REQ-M01/14; FMEA	G2
R-02	Comfort not demonstrated / wrong stiffness	High	SIM-001 multi-band	G2
R-03	Lock fail / false SPORT	High	Confirm sensor; REQ-M17	G3
R-03C	COMFORT roll/handling in manoeuvre	High	REQ-M21; anti-roll study	G2
R-04	Hydraulic leak / fire	High	Detect; fluid selection	G3
R-04S	S4 soft handling at speed	High	Lock preference; limp-home ≤80 km/h class	G2
R-05	Module mass / thermal / supply	High	Datasheet; dual source	G3
R-08	Stop energy insufficient	Cat.	REQ-M09; STR-001	G3
R-10	EMC / cabin field	High	Field map; ≤0.5 mT plan	G5
R-11	Gap/lock contamination	High	Seals; RPN 112	G3
R-12	Prior art on predictive modes	Med.	Counsel search before broad claims	G0
R-13	ICD / liability chain	High	Contract + ICD	G1
R-14	Power budget / 48 V peaks	Med.	Close budget at L2	G3
R-15	Cost / schedule	High	Phased PoC	G1
R-16	FTO / chain of title	High	Counsel; assignment log	G0
R-17	Export control / sanctions compliance	High	Legal screen per counterparty	G0
R-18	Regulatory category / speed limiter	High	REG-001 matrix	G2

16. Disclosure Tiers and NDA Rules

Tier	Content	Condition
T0	Teaser without enabling detail	Public
T1	This Document 00	Executed NDA

Tier	Content	Condition
T2	Models, ICD, simulation, commercial terms	NDA + need-to-know
T3	Claim charts / IP-001	Counsel-controlled

17. Architecture Focus Themes and Prior-Art Caution

Focus themes: capture invariant; macro–micro split; dual-mode COMFORT/SPORT; module as Path A product; honest OPEN items. **Finding 19:** Predictive radar/GPS mode selection and lockable cabin suspensions are crowded prior-art areas. Differentiation for counsel should emphasise mechanical lock, defined stiffness constants, module ICD and capture invariants — not mode-select user experience alone. Search before broad enabling disclosure. No novelty claim is made here.

18. Explicit Non-Claims

- Any measured ride-comfort improvement percentage without a configuration-managed SIM/test package.
- Any certification, type-approval or regulatory compliance.
- Any crashworthiness or survivability demonstration.
- Any patent grant or freedom-to-operate.
- Any production-ready mass, cost or schedule commitment.
- Any continuous public-road speed capability at 250 km/h.
- Any closed mass or energy budget suitable for production release.
- Any clinical outcome in ambulance use.
- Any continuous RMS force capability until thermal data exist.
- Any supplier or OEM endorsement.
- Any verified Km or continuous power figure.
- Any moose-test or handling ranking versus production vehicles until test data exist.
- Free magnetic levitation or marine flotation capability.

19. Immediate Engineering Work Package

Pri	Work product	Owner	Gate
1	SIM-001 multi-band with declared passive k, c and road classes	Chassis	G2
1	Numeric COMFORT roll / lateral-g policy + anti-roll study	Dynamics	G2
1	S4 HARA row + limp-home speed policy	Safety	G2
1	STR-001 stop energy and stroke	Structures	G3
1	Mass book rebuild including fuel system and corner masses	Integration	G1
2	Hydraulic sizing + total module power budget	Module	G3
2	Seals/bellows + contamination detection	Module	G3
2	REG-001 matrix; cabin field map plan	Regulatory / EMC	G5
2	Prior-art search before expanded mode claims	IP	G0

20. Glossary (selected)

Term	Meaning
CMFSB	Captured Mechanical Floating Safety Body
Floating	Body isolation relative to chassis — not free levitation, not marine flotation
COMFORT / SPORT	Soft isolation mode / locked handling mode
Reference path	Hybrid hydraulic + Lorentz
OPTION path	PM+EM requiring stability gate
L0–L4	Evidence ladder levels (§14)
Ph / G / Pri	Phase / Gate / Priority — exclusive naming
Line-sum P90	Sum of per-line P90 masses — conservative versus joint statistical P90

22. Dual-Mode Architecture — COMFORT and SPORT

Dual-mode operation is a primary R4.8 addition retained in R4.9. COMFORT provides soft isolation across the full macro envelope with active micro control. SPORT engages body locks so the upper body seats on the lower body and effective roll/yaw stiffness rises toward rigid-body class. Mode selection may follow driver command, radar-based road sensing, or map–GPS arbitration with a frozen priority table.

Correction (Finding 3) — COMFORT handling risk: Soft corner support reduces roll stiffness. Illustrative estimate: $k \approx 40$ kN/m per corner, lateral tower spacing 1.4 m, CG about 0.6 m above roll centre → about 4.6° roll at 0.4 g (about 11°/g class), higher than typical passenger cars. A mode transition of ≤ 3 s is slower than many emergency manoeuvre timescales (for example a moose test). Required controls: (1) numeric COMFORT roll-gradient limit (°/g) agreed with the vehicle dynamics owner; (2) anti-roll evaluation — hydraulic cross-link and/or active anti-roll in COMFORT; (3) numeric lateral-g inhibit threshold with default toward SPORT/lock above threshold; (4) speed-dependent default mode policy. Do not rely on transition time alone for emergency manoeuvres.

22.1 Mode definitions

COMFORT mode: locks are released; the macro stage is free to use its design stroke; the micro stage regulates gap and applies dynamic force within authority limits; the control objective is isolation under the multi-band rules of §24. COMFORT does not mean the body is free-floating; capture geometry remains engaged.

SPORT mode: locks engage so that the upper body seats on the lower body or lock faces; relative macro motion is intentionally constrained; handling-related roll and pitch stiffness rise; isolation authority may be reduced or inhibited depending on the frozen control calibration. SPORT is a handling mode, not a crash-mode substitute for progressive stops.

22.2 Mode comparison matrix

Attribute	COMFORT	SPORT
Lock state	Released / confirmed open	Engaged / confirmed closed
Macro stroke available	Full ± 50 mm TARGET envelope	Constrained by lock geometry
Primary objective	Isolation (multi-band metrics)	Handling stiffness / predictability
Magnetic/Lorentz force	Active within limits	May be reduced or hold-only per calibration
Transition time TARGET	Into COMFORT ≤ 3 s class after unlock confirm	Into SPORT ≤ 3 s class after lock confirm
Crash path	Stops + structure	Stops + structure (lock not primary)
Power-loss default interaction	S4 passive path; lock preference per §8	If already locked, remains constrained subject to fail rules

22.3 Lock and release architecture (ATFSV-LCK-001 planning)

Lock requirement	Statement	Status
State sensing	Independent confirm of engaged / released	TARGET
Engage time	≤ 3 s TARGET including confirm	TARGET
Release interlock	Inhibit release if capture/stroke not nominal	TARGET
Contamination	Sealed interface; detect degraded confirm	TARGET
S4 command	Prefer engage when residual energy allows	BASELINE preference
Crash role	Not primary energy absorber	BASELINE

22.4 Three-path arbitration

Driver command, radar-based surface/event detection, and map–GPS policy are three input paths. A frozen priority table shall define override order (for example: safety inhibit > driver SPORT request > map policy > comfort default). Radar/GPS paths are TARGET convenience features; they are not safety mitigations for COMFORT roll risk and do not replace numeric lateral-g / roll-gradient limits. Prior-art caution in §17 applies before broad disclosure of predictive logic.

Path	Typical use	Limitation
Driver	Explicit COMFORT/SPORT request	May be overridden by safety inhibit
Radar-based	Anticipatory mode suggestion / inhibit	Sensor availability; not sole safety case
Map–GPS	Road-class policy defaults	Map quality; localization error

22.5 Transition timing and integrity

Transitions shall never pass through unconstrained free-float. Lock release requires capture and stroke conditions to be nominal. False SPORT (claiming lock when not confirmed) is a high-severity failure (R-03). False COMFORT under high lateral demand is addressed by R-03C and REQ-M21.

22.6 Control and hydraulic implications

COMFORT demands compliant macro behaviour and controlled leveling. SPORT demands rapid, confirmed lock-up and may require isolation of certain compliant paths. Valve fail states on power loss remain fail-closed for height retention as stated in §8. Flow demand for four-corner settle within ≤3 s shall be shown in the hydraulic budget of §4 before L2.

22.7 Attractiveness note (non-binding)

Dual-mode is commercially attractive because it avoids forcing a permanent soft-only compromise on vehicles that sometimes need crisp handling. It is not, by itself, evidence of comfort performance. Comfort evidence remains tied to §24 and the evidence ladder.

23. Path A Commercial Option Framework (Planning — Non-Binding)

Correction (Finding 21): Country-named counterparties and any USD price discussion are removed from this technical baseline. International counterparties may be engaged under NDA subject to applicable export-control and sanctions screening in the relevant jurisdictions; legal counsel screens each counterparty. Option fees, license rates and purchase prices live only in T2/T3 commercial packages, not in Doc 00.

Item	Doc 00 statement
Path A object	Field-limited module license and/or supply of ATFSV-MOD-001
Structure (indicative)	Evaluation option → term sheet with upfront + milestones + royalty — commercial numbers outside Doc 00
Not delivered by Path A alone	Full vehicle certification, Path B carbon-cell programme, or guaranteed comfort percentage
Buyer actions that help	Fund L1/L2 evidence; freeze field definition; complete title assignment chain

23.1 Engagement ladder (recommended sequence)

T0 teaser → NDA → T1 Doc 00 review → technical questions log → optional evaluation license → L1/L2 evidence review → term sheet discussion under T2. Skipping evidence and jumping to price discussion usually destroys trust on both sides.

24. ATFSV-SIM-001 — Simulation Skeleton (Comfort versus Sport)

Purpose: quantify isolation value honestly and prevent mis-assigning magnet-carriage stiffness to body bounce frequency. Minimum model class: quarter-car or half-car. Three-mass screening is acceptable only for PRELIMINARY orientation runs and is not verification evidence.

Required input	Rule
Passive reference k, c	Body-support values for ~1.0–1.4 Hz class — published in the run record
Road profiles	ISO 8608 class B and class C at declared speeds (or measured profiles)
Payload cases	Including empty and 4×100 kg class
Mode cases	COMFORT and SPORT (locked) as applicable
Speed cases	At least the speeds associated with the chosen road classes
Required output	Rule
Frequency bands	0.1–0.5 / 0.5–2 / 4–12.5 Hz reported separately
Minima	Per-band minimum TARGET frozen in the SIM plan — not a single ≥10% only
Force and travel	Actuator force and relative travel histograms; compare to ±2.5 kN and ±50 mm
Power	Mean isolation power for ATFSV-ENE-001

Screening PRELIMINARY results under an idealised linear three-mass model (not proof): passive stage about 66% weighted-RMS reduction versus rigid cabin; active layer about 35% further versus that passive stage; 3σ force about 0.4–0.65 kN; 3σ relative travel about 17–26 mm. These numbers orient force and stroke budgets; they do not authorise marketing claims.

24.1 Execution checklist before calling a run complete

Record software version, parameter set, road file identifiers, payload, mode, and the exact passive reference k,c. Store raw time histories sufficient to regenerate band metrics. Mark every plot PRELIMINARY until independent review is logged.

25. Path B Structural Topics — Summary Only

Correction (Finding 20): Detailed six-occupant carbon cell, FIA-referenced roll structure, dual 4×4/6×6 running gear, and stretch mass ceiling narrative are OUT OF BASELINE for Path A module licensing and are maintained in ATFSV-DOC-01. Brand comparison names are removed from Doc 00. Only the summary table below remains so that Path B is acknowledged without diluting the module-primary purpose of Document 00.

Topic	Summary in Doc 00	Detail location
Carbon passenger cell	Path B study; mass OPEN	ATFSV-DOC-01
Roll structure	Path B study	ATFSV-DOC-01
4×4 / 6×6	Path B packaging options	ATFSV-DOC-01
Mass ceiling below 9,000 kg	Stretch study only — not operational baseline	OUT OF BASELINE
Operational GVW	7,500 kg single baseline	Appendix A

Appendix A — Reference Vehicle (Path B context only)

Appendix A supports Path B discussion and programme accounting. Path A module evaluation does not require acceptance of Appendix A powertrain or range targets. All status labels remain as defined in §0.1.

A.1 Configurations

Item	Operational baseline	Development
GVW	7,500 kg	6,000 kg lab/mule
Purpose	Operational accounting, range, brake, tyre, payload	Lab / mule only
Peak power / speed	3,000 hp gated; lower-speed certification path may be selected	Gated development only

A.2 Mass lines (corrected)

Corrections (Findings 6–7): Single operational GVW baseline = 7,500 kg. The stretch ceiling below 9,000 kg is OUT OF BASELINE. Corner wheel+tyre+brake for six high-rated corners is rebuilt to about 560/640 kg (was 400/480). Fuel system dry (tank, pump, lines) is added at about 40/55 kg. Illustrative RSS uncertainty on curb elements about +373 kg yields a statistical P90 curb about 6,576 kg and payload about 924 kg (PRELIMINARY until ATFSV-MASS-001). Line-sum P90 is more conservative and can erase payload — active mass control is mandatory. Carbon cell mass remains OPEN under Path B.

WBS line	P50 kg	Line-sum P90 kg	Notes
Curb allocation (programme STUDY)	5,850	6,940	R4.8 study retained
Occupants 4×100	400	400	Payload — not empty curb
Fuel system dry (tank, pump, lines)	40	55	ADDED
Wheel + tyre + brake (six corners)	560	640	REBUILT (was 400/480)
Carbon cell structure	OPEN	OPEN	Path B / DOC-01

A.3 Mission energy and brake (7,500 kg basis)

Correction (Finding 8): Road load 81.6 kW at 120 km/h gives about 816 kWh mechanical over 1,200 km before auxiliaries. The R4.8 figure ≈886 kWh implies about +7 kW auxiliaries or a higher mean speed near 128 km/h — the choice must be stated explicitly in ATFSV-ENE-001. Drivetrain efficiency 28–30% STUDY unless a supplier map replaces it. Without large auxiliaries at 120 km/h, fuel with 10% reserve is about 336–360 L; ≥390 L remains the conservative TARGET.

Quantity	Screening value	Note
Rolling power @ 120 km/h	≈29.4 kW	DERIVED
Aero power @ 120 km/h (CdA 2.3 m²)	≈52.2 kW	STUDY
Road load @ 120 km/h	≈81.6 kW	Before auxiliaries
Road energy 1,200 km @ 120 km/h	≈816 kWh	Before aux
Mission energy line published in R4.8	≈886 kWh	Reconcile aux or speed in ENE-001
Efficiency to fuel	28–30% STUDY	Unless supplier map
Fuel with 10% reserve	~336–360 L at 120 km/h without large aux; TARGET ≥390 L conservative	TARGET
Single stop from 250 km/h	≈18.1 MJ	Kinetic screening
Repeated cycle 250/200/150/120	≈40 MJ cumulative class without regen	Fade risk

Brake thermal model (ATFSV-BRK-001) and full energy chain (ATFSV-ENE-001) remain OPEN. 250 km/h is gated development only — not a continuous public-road claim.

26. Master Position Statement

ATFSV-MOD-001 is a mechanically captured, dual-mode (COMFORT/SPORT) isolation module intended primarily for Path A license and supply. R4.9 corrects quantitative and structural issues identified against R4.8 while preserving structured tables and engineering explanations from the R4.8 module-primary baseline. Path B vehicle detail is summarised only and lives in ATFSV-DOC-01. Evidence remains ladder-based. Engineering maturity is TRL 2–3. This document is not a certificate of conformity and does not assert patentability or freedom-to-operate.

Buyers and partners should treat every TARGET and STUDY value as a planning number until the linked gate and evidence level are closed. The strongest near-term diligence products are ATFSV-SIM-001 (multi-band), a single-tower force–gap curve, and a written S4 handling policy. Module ICD release is the main contractual interface for Path A supply discussions.

Appendix B — Revision History, Change Log and Distribution

Rev	Date	Summary		
4.8	2026-10-02	Module-primary dual-mode external baseline		
4.9	2026-10-02	R4.8 body and tables retained; Findings 1–23 corrected: 3g loads, gap-dependent PM bias, multi-band SIM + PRELIMINARY screening, COMFORT handling, S4 lock/limp-home, numeric crash stop, GVW/mass/fuel/corner rebuild, mission energy reconcile, power/hydraulic budgets, FMEA RPN, REQ/risk/gate trace, exclusive vocabulary, regulatory matrix, cabin field plan, desensitised commercial section, Path B summary only, drawn Figures 1–4, position statement at end; dense continuous layout		
	Role	Name / organisation	Date	Signature
	Author	ATFSV Engineering	2026-10-02	
	Reviewer	TBD		
	Approver	TBD		

Appendix C — Corrections Index (Findings 1–23)

ID	Topic	Disposition in R4.9
1	Figures	Drawn Figures 1–4 (layers, state, power-loss)
2	SIM multi-band	§0.2 and §24; PRELIMINARY screening published
3	COMFORT handling	§22 numeric roll/lat-g/anti-roll/speed policy
4	S4 at speed	§8 lock preference, limp-home, HARA row
5	Crash stop	REQ-M09a/b/c numeric
6	GVW	Single 7,500 kg baseline; <9 t OUT OF BASELINE
7	Mass lines	Corner 560/640; fuel 40/55
8	Mission energy	816 vs 886 reconciled; efficiency 28–30%
9	3g loads	S 9–15; M 15–21; L 21–24 kN
10	Power/hydraulic	§4 full budget tables
11	FMEA RPN	§9 RPN column + seals + lock rows
12	PM bias	Gap-dependent at 15 mm
13	REQ trace	§7 risk/ICD/gate
14	Risks	§15 expanded register
15	Naming	Ph / G / Pri / L
16	Vocabulary	§0.1 exclusive labels
17	Standards	§12.1 matrix
18	Magnetic limit	≤0.5 mT planning
19	Prior art	§17 counsel caution
20	Path B scope	§25 summary; DOC-01
21	Commercial	§23 desensitised
22	Structure	Position at end; change log; TOC
23	PDF quality	Metadata title; continuous dense tables and prose

END OF DOCUMENT — ATFSV-DOC-00-R4.9

Preliminary Engineering Baseline • Not Certified • TRL 2–3 • Confidential IP

Annex E — Reference Application A and Programme Context (from R4.8)

Lead application proposals remain flexible: specialty/defence utility, luxury limited-series, or patient-transport cabin classes. Cabin mass class 1,500–2,500 kg study remains the Path A sizing context for upper-body share. Reference Application A is a screening application only and does not freeze a launch market. Buyer-specific standards (e.g. ambulance EN 1789 where applicable) are confirmed by the buyer and are not claimed as baseline compliance in this document.

Programme maturity remains TRL 2–3. Evidence ladder L0–L4 defines what may be stated externally. Comfort percentage language is forbidden until L1 multi-band SIM and preferably L2 bench data exist under configuration management.

Context	R4.9 statement	Status
Cabin mass class	1,500–2,500 kg study for upper-body share	STUDY
Lead applications	Specialty / defence utility / luxury limited / patient transport — flexible	PROPOSED
TRL	2–3	BASELINE honesty
Comfort claims	Forbidden until L1 (prefer L2)	BASELINE rule

Annex F — Lorentz Screening Numbers (R4.8 retained as STUDY)

R4.8 published screening Lorentz continuous force and copper-loss discussion. R4.9 retains these as STUDY and explicitly states they are not total module power. Continuous force capability depends on thermal path, duty cycle and ambient. Peak ±2.5 kN remains STUDY until bench confirmation. RMS continuous ≈0.7 kN planning may be oversized relative to road-noise isolation need identified in multi-band screening; leveling and manoeuvre loads are separate sizing cases.

Quantity	Screening value	Status
Peak dynamic force	±2.5 kN per tower	STUDY
RMS continuous planning	≈0.7 kN RMS	STUDY — may be oversized for road noise
Copper loss discussion	≤0.5 kW class partial	STUDY — not total power
Bandwidth	≥20 Hz TARGET	TARGET
Gap range	15–30 mm	TARGET
Micro stroke	±7.5 mm	TARGET

Annex G — Per-Corner Mass Breakdown (STUDY screening, R4.8 structure)

R4.8 included per-corner screening mass elements. R4.9 retains the structure and points vehicle-level rebuilt lines to Appendix A (corner 560/640 kg for six high-rated corners; fuel 40/55 kg). Module core remains ≤55 kg TARGET independent of unsprung corner share.

Corner element (STUDY)	Role	Note
Core tower structure	Primary module mass	Within ≤55 kg TARGET
Lorentz assembly	Coils, magnets, local mount	Within TARGET unless ICD splits
Local sensors / harness	Gap, stroke, temp	Within TARGET unless body-side
Hydraulic interface share	Local valves/lines at tower	STUDY — ICD boundary
Unsprung wheel/tyre/brake share	Vehicle mass book	Appendix A rebuild 560/640 for six

Annex H — Commercial Milestone Sketch (R4.8 §23 retained, prices removed)

R4.8 included Path A option and milestone language. R4.9 retains the structure and removes any USD figures and country-named counterparties (Finding 21). Commercial numbers live only in T2/T3 packs under NDA.

Stage	What happens	Evidence expectation
Option / evaluation	Time-limited study rights under NDA	T1 Doc 00 sufficient to start
Term sheet	Field definition, exclusivity, milestones	L1 preferred; L2 strengthens
Milestone 1	ICD freeze + SIM package	G1–G2
Milestone 2	Single-tower bench (force, power-loss, lock)	L2 / G3
Milestone 3	Four-tower / HIL	L3 / G4
Supply / royalty phase	As contracted after milestones	Outside Doc 00

What Path A delivers: module architecture rights/supply within contracted field; dual-mode SPORT/COMFORT capability as designed; ICD and safety behaviour as frozen. What Path A does not deliver alone: full vehicle certification; Path B carbon-cell programme; guaranteed comfort percentage; production cost or royalty figures in this technical baseline.

Annex I — Buyer Actions That Shorten Time-to-Term-Sheet (R4.8)

- Execute NDA promptly and nominate a single technical counterpart for the questions log.
- Fund or co-fund L1 multi-band SIM with agreed passive reference k,c and road classes.
- Fund or co-fund L2 single-tower force–gap and power-loss demonstration.
- Freeze field-of-use definition early (vehicle classes, territories, exclusivity).
- Complete inventor assignment and chain-of-title documentation before term sheet.
- Accept OPEN items as gated work rather than as reasons to reject the architecture a priori.
- Keep comfort marketing language off the table until evidence ladder allows it.

Annex J — Path B Non-Claims Specific to Structural Options (R4.8 §25.6 adapted)

Even in ATFSV-DOC-01, the following remain non-claims until evidence exists: (1) any specific carbon cell mass; (2) any FIA homologation; (3) any 4×4/6×6 production readiness; (4) any mass ceiling below 9,000 kg as operational baseline; (5) any brand comparison. In Document 00 these topics appear only as summary pointers (Finding 20).

Annex K — ICD Mandatory Headings (Path A Interface)

ICD section	Minimum content	Owner
Mechanical	Bolt pattern, stiffness targets, stop faces, service envelope, sealing surfaces, mass properties	Structures/Chassis
Hydraulic	Ports, pressure class, fluid, fail positions, cleanliness	Module
Electrical	48 V limits, connectors, shielding, ground	EE
Communications	Bus type TBD, latency, CRC, fail-silent vs fail-indicate	Controls
Safety inhibits	Lines, power-loss defaults, prohibited states	Safety
Lock	Engage/release confirms, interlocks, contamination	Lock systems
Environment	Temperature, sealing IP target, corrosion, shock	Module

ICD release is Gate G1 content. Without it, Path A term sheets lack a stable interface and change cost becomes unbounded.

Annex L — COMFORT Manoeuvre Worked Example (Finding 3 detail)

Assume four corners at k = 40 kN/m in COMFORT, lateral spacing between left and right support points L = 1.4 m, and roll centre to CG height h = 0.6 m. Under a steady 0.4 g lateral acceleration, roll angle of order 4–5° can appear (about 10–12 °/g class). Many passenger cars target lower steady-state roll gradients. Therefore COMFORT without anti-roll assistance or mode inhibit can feel soft in lane-change and roundabout events even if straight-road isolation is attractive.

Parameter	Illustrative value	Role
k per corner	40 kN/m	COMFORT support stiffness

Parameter	Illustrative value	Role
Track of supports L	1.4 m	Roll stiffness geometry
h (CG above RC)	0.6 m	Roll moment arm
ay	0.4 g	Steady lateral example
Resulting roll (order)	~4.6° (~11 °/g)	Why limits are required

Design responses inside the architecture: increase COMFORT roll stiffness via hydraulic cross-link; add active anti-roll moments within actuator authority; lower the lateral-g threshold at which SPORT or partial lock is commanded; bias default mode with speed. Transition time ≤ 3 s alone does not solve a 0.5–1.0 s emergency lane-change. REQ-M21 forces these choices to become numeric rather than rhetorical.

Annex M — S4 Handling Narrative for Safety Review (Finding 4 detail)

Safety reviewers will ask what the driver feels if 48 V is lost at highway speed in COMFORT. R4.9 answers in three layers. Layer 1 — physics: magnetic force drops; valves fail-closed; body sits on passive support. Layer 2 — policy: prefer lock engagement if energy allows; otherwise limp-home speed TARGET ≤ 80 km/h class and driver indication. Layer 3 — process: HARA row, FMEA line, and test cases in L2/L3.

S4 case	Intended behaviour	Evidence
COMFORT + power loss + lock energy available	Command lock; confirm; indicate degraded state	Bench + HIL
COMFORT + power loss + no lock energy	Passive support + speed limit request	Policy + driver HMI
SPORT already locked + power loss	Remain locked; inhibit false unlock	Interlock test
Return of power	S4→S1 self-test only; no direct S4→S2	State-machine test

Annex N — SIM Reporting Template (minimum columns)

Column	Content
Run ID / date / software	Unique identifier and configuration hash
Passive k, c	Numeric body-support values and mass used
Road file / class / speed	ISO 8608 class or measured profile ID + speed
Payload / mode	Mass case and COMFORT or SPORT
Band metrics	RMS or equivalent in 0.1–0.5 / 0.5–2 / 4–12.5 Hz
Force / travel peaks	Including 3 σ or percentile as defined in the plan
Mean power	For energy coupling
Marking	PRELIMINARY or later CM status — never silent

Annex O — Diligence Reading Guide

Recommended sequence for an external technical reader: (1) datasheet and invariants; (2) §8 state machine and S4 decision; (3) §7 requirements and §15 risks; (4) §24 SIM rules; (5) Appendix A only if Path B is in scope; (6) Appendix C checklist against the buyer’s own review notes.

When a row is marked OPEN, the correct buyer question is not “is the architecture invalid?” but “who owns the closure and at which gate?” When a row is STUDY, the number may change without a formal baseline break until promoted to TARGET or BASELINE. When a row is PRELIMINARY, it may be used for orientation and test planning only — not for marketing performance claims.

Annex P — Configuration Management Preview

R4.9 is not yet under full configuration management. When CM starts, the following artefacts should receive identifiers and owners: ATFSV-DOC-00 (this baseline), ATFSV-MOD-001 (module datasheet), ATFSV-ICD-001, ATFSV-SIM-001, ATFSV-STR-001, ATFSV-LCK-001, ATFSV-MASS-001, ATFSV-ENE-001, ATFSV-BRK-001, ATFSV-REG-001, ATFSV-IP-001. Change requests shall reference requirement IDs and risk IDs when altering BASELINE content.

Until CM is active, PDF date and revision string in the footer are the only practical identity. Recipients shall not mix pages from R4.8 and R4.9 in a single binder without marking superseded pages.

Annex Q — Full Dual-Mode Requirements Cross-Reference (R4.8 §22.7)

The dual-mode architecture maps to the module requirements register as follows. This cross-reference is retained from R4.8 and updated for R4.9 requirement IDs.

Dual-mode topic	REQ IDs	Risk IDs	Gate
SPORT lock engage/confirm	M17, M19	R-03	G3
COMFORT full envelope	M18	R-03C	G3
No free-float transition	M20, M01	R-01	G2
COMFORT roll/lat-g policy	M21	R-03C	G2
Arbitration priority table	M19	R-12	G3
S4 lock preference	M10	R-04S	G2
Seals / contamination at lock	M23	R-11	G3

Control calibration for SPORT may reduce Lorentz authority once lock is confirmed, because handling stiffness is then dominated by the mechanical path. COMFORT calibration uses the full force budget within thermal and stroke limits. Mode transition shall not command force that exceeds stop design limits.

Annex R — Architecture Focus Themes Table (R4.8 §17 retained)

Theme	Description
Macro–micro kinematic split	Large body travel on passive/compliant stage with local high-bandwidth force stage
Fail-safe capture under power loss	Transition to mechanical/hydraulic support without software aliveness; fail-closed leveling
Progressive-stop energy path	Ultimate crash energy absorption independent of closed-loop magnetic force
Leveling + isolation co-architecture	Active leveling that preserves dynamic isolation stroke under payload change
Modular tower ICD + safety behaviour	Interface-controlled isolation module with defined force, stroke, bandwidth and fail-safe rules
State-machine physical preconditions	Deterministic transitions requiring physical confirmation before recovery or high-force enable
Dual-mode SPORT / COMFORT	Rigid seated lock for handling vs full isolation for soft ride; three-path selection

These themes describe architectural focus for technical understanding. They are not asserted patent claims and do not constitute a patent disclosure. Detailed claim charts remain in T3 (ATFSV-IP-001).

Annex S — Timeline Acceptance Detail (R4.8 §13.1 expanded)

Ph0 acceptance requires: geometric clash check through full stroke; lock face clearance in both modes; draft ICD reviewed by mechanical, electrical and safety owners. Ph1 acceptance requires: measured force–gap or force–current curve; demonstrated passive landing on power loss; thermal RMS within study limits; lock engage and release with positive confirmation. Ph2 acceptance requires: four-corner 6-DOF behaviour on representative mass; SPORT↔COMFORT within ≤3 s TARGET under defined conditions; power-loss from each mode; stop impact within design envelope. Ph3 (optional) acceptance requires: structured ISO 2631-1 comparison and handling evaluation on a donor chassis.

Phase	Typical exit review questions
Ph0	Does the envelope close? Is ICD draft usable for supplier RFQ?
Ph1	Does power-loss land safely? Is force curve credible? Does lock confirm?
Ph2	Do four corners stay coherent? Are mode transitions deterministic?
Ph3	Does mule data support L4 discussion? What remains OPEN?

Annex T — Hydraulic Fail Positions and Cleanliness (planning)

Fail-closed leveling valves on power loss are BASELINE. Accumulator sizing must support leak-down and settle pulses. Fluid selection affects fire risk, environmental compliance and seal life — OPEN until FMEA and ICD freeze. Cleanliness targets (e.g. ISO 4406 class) shall appear in the ICD hydraulic section before supplier RFQ. Hose burst is a high-severity FMEA item; progressive stops must carry residual load if pressure is lost.

Hydraulic topic	R4.9 position	Closure
Leveling valve fail state	Fail-closed BASELINE	ICD + test
Accumulator	OPEN sizing	Calc note + L2
Fluid	OPEN selection	FMEA + ICD
Cleanliness	OPEN class	ICD + process
Burst residual path	Stops carry load	STR-001 + FMEA

Annex U — Electromagnetic and Cabin Field Planning (Finding 18)

Cabin magnetic field planning target ≤0.5 mT at seating positions is a planning value pending formal guidance confirmation for the target market. It is not a verified measurement. Field mapping is G5-linked work. Shielding, routing and magnet orientation are design levers. Implantable device considerations, if any, follow applicable guidance once the market and vehicle category are frozen.

Topic	Statement	Status
Planning limit at seats	≤0.5 mT	TARGET plan
Measurement plan	Map under max continuous and peak duty	OPEN
Guidance confirmation	Market-specific formal reference	OPEN
Design levers	Shielding, routing, orientation	STUDY

Annex V — Export Control and Counterparty Screening (Finding 21)

International counterparties may be engaged under NDA subject to applicable export-control and sanctions screening in the relevant jurisdictions. Legal counsel screens each counterparty. Document 00 does not name countries or price points. Technology classification (if any) is outside this baseline and must be determined under applicable law before controlled technical data transfer beyond T1.

Risk R-17 tracks this item at Gate G0. Failure to screen is a programme risk, not a technical architecture flaw. Buyers in regulated sectors should state their compliance constraints early so that disclosure tiers can be adjusted.

Annex W — Statistical Mass Method Note (Findings 6–7 detail)

Operational GVW = 7,500 kg is the only baseline number for payload arithmetic in Doc 00. Using line-sum P90 curb plus occupants without statistical correlation double-counts conservatism and can show near-zero payload even when a joint statistical P90 would not. R4.9 reports both the rebuilt line items and an illustrative RSS view, and requires ATFSV-MASS-001 to freeze the method. Until then, programme management shall treat payload as OPEN for sales commitments.

View	Curb class	Payload class at 7,500 kg	Status
P50 style	~5,850 kg study	~1,650 kg before options	STUDY
Line-sum P90 style	Can exceed 7,100 kg with occupants	Can collapse toward low hundreds of kg	Conservative warning
Illustrative RSS	~6,576 kg	~924 kg	PRELIMINARY

Annex X — Brake Thermal Screening Context (Appendix A)

Single stop kinetic energy from 250 km/h at 7,500 kg is about 18.1 MJ. A repeated high-speed cycle without regeneration accumulates on the order of 40 MJ class and raises fade risk. ATFSV-BRK-001 remains OPEN. 250 km/h is gated development only and is not a continuous public-road claim. Path A module evaluation does not require acceptance of the brake thermal model.

Annex Y — Explicit Mapping of R4.8 Section Numbers to R4.9

Annex AA — Complete Non-Claims List (R4.8 §18 expanded)

This document does not claim or assert any of the following. Buyers and partners must perform independent due diligence.

- (1) Any measured ride-comfort improvement percentage without a configuration-managed SIM/test package.
- (2) Any certification, type-approval or regulatory compliance.
- (3) Any crashworthiness or survivability demonstration.
- (4) Any patent grant or freedom-to-operate.
- (5) Any production-ready mass, cost or schedule commitment.
- (6) Any continuous public-road speed capability at 250 km/h.
- (7) Any closed mass or energy budget suitable for production release.
- (8) Any clinical outcome in ambulance use.
- (9) Any continuous RMS force capability until thermal data exist.
- (10) Any supplier or OEM endorsement.
- (11) Any verified Km or continuous power figure.
- (12) Any moose-test or handling ranking versus production vehicles until test data exist.
- (13) Free magnetic levitation or marine flotation capability.
- (14) Any VERIFIED performance metric under configuration management at this revision.
- (15) Any guaranteed dual-mode transition performance outside defined test conditions.

Annex AB — Module Boundary vs Vehicle Boundary (clarity)

Path A commercial and technical discussions stop at the module ICD. Vehicle-level GVW, powertrain, fuel volume, brake thermal model and category/speed regulation are REFERENCE context in Appendix A and do not have to be accepted for a Path A evaluation. Conversely, Path B negotiations must open mass, energy, brakes, body structure (DOC-01) and REG-001. Mixing the two without a document split is a common source of delay.

Topic	Path A need?	Path B need?	Where documented
Module force/stroke/mass	Yes	Yes	§5–§7
S4 / state machine	Yes	Yes	§8
SIM multi-band	Yes (comfort claims)	Yes	§24
GVW 7,500 kg	No	Yes	App A
Fuel / range	No	Yes	App A
Carbon cell / roll	No	Yes	DOC-01
REG-001 category	Limited	Yes	§12 / REG-001

Annex AC — Progressive Stop Design Intent (Finding 5 expanded)

Progressive stops are the ultimate energy path when free play is consumed. Design intent: (1) force–displacement curve rises smoothly to avoid peak structural shock; (2) stroke after free play is sufficient for the energy of credible events; (3) distribution across corners accounts for asymmetric engagement; (4) materials and temperature derating are documented; (5) inspection intervals appear in service documents. STR-001 owns the curve before G3.

Stop requirement	R4.9 numeric / rule	Owner
Stroke beyond free play	≥50 mm or equivalent energy	STR-001
Energy screening order	~22 kJ at 2,250 kg class	STR-001
Force ceiling screening	~441 kN total at 20 g × 2,250 kg class	STR-001
S5 free-play reduction	Lock and/or hydraulic close on precursor	Safety + Module
Inspection	Interval in service plan	Service

Annex AD — Leveling Performance and Stroke Margin

At k ≈ 40 kN/m and ≈562 kg corner share, static deflection under 1 g is ≈25 mm. Payload change of 100 kg per corner adds ≈25 mm. Without leveling, available dynamic stroke shrinks. Four-corner settle ≤3 s TARGET restores ride height. Leveling rate limits avoid exciting body modes. Leveling is not a substitute for isolation control and is rate-limited so it does not fight the micro stage at isolation frequencies.

Quantity	Screening value	Implication
Corner mass share	~562 kg at 2,250 kg body	f_n sizing
k body-support	~40 kN/m example	~1.34 Hz class
Static 1 g deflection	~25 mm	Stroke budget
100 kg payload delta	~25 mm additional	Leveling required
Settle TARGET	≤3 s four-corner	Pump/accumulator size

Annex AE — Sensor and Confirm Strategy (planning)

Minimum sensor set for baseline safety behaviour includes: gap or relative stroke at each tower; lock engage/release confirms; hydraulic pressure (or equivalent support-health signal); module supply voltage; temperature at coil or critical thermal node. Optional: cabin accelerometers for monitoring; radar/GPS inputs for mode suggestion only. Confirm signals for lock are independent of the command line. Loss of confirm inhibits SPORT claim (false SPORT prevention).

Signal	Use	Fail behaviour
Gap / stroke	Control + stop proximity	Enter S3 / bound force
Lock confirm	SPORT validity	Inhibit S2S claim
Pressure / support health	Static path integrity	S3 / limp
Supply voltage	S4 entry	Hardware path to S4
Temperature	Thermal protect	Derate / S3

Annex AF — Service, Rescue and Maintenance Access

REQ-M15 requires manual recovery and rescue access provisions. Design shall allow: (1) mechanical isolation of high force without defeating capture in an uncontrolled way; (2) access to lock release under emergency procedures; (3) maintenance lockout (S8) with physical device; (4) documentation for first responders where vehicle integration applies (Path B). Module-level service manuals are Path A deliverables after ICD freeze.

Annex AG — Summary of What Changed from R4.8 to R4.9 (narrative)

R4.9 does not rewrite the architecture. It keeps the captured macro–micro dual-mode module, the S0–S8 state machine, the Path A primary commercial path, and the evidence ladder. It corrects quantitative inconsistencies (3 g bands, PM bias gap dependence, corner and fuel mass, mission energy reconciliation), strengthens traceability (REQ risk/gate, FMEA RPN, expanded risks), adds multi-band SIM rules and PRELIMINARY screening results, adds COMFORT handling and S4-at-speed policies, numeric crash stop requirements, drawn figures, exclusive vocabulary and naming governance, regulatory matrix depth, cabin field planning target, desensitised commercial section, and Path B summary-only scope. Position statement is at the end. Change log holds revision storytelling.

Content volume is increased relative to a thin correction sheet so that R4.8 explanations are not lost. Page count at dense formatting is at least sixteen pages without intentional blank pages.

Annex AH — Detailed REQ Rationale Notes (preserving R4.8 engineering intent)

REQ-M01/M02/M14/M20 exist because capture and passive support are the public safety story of CMFSB. Without them, the product collapses into a conventional active mount discussion and loses its architectural differentiator.

REQ-M08/M21 exist because comfort value and handling risk are coupled. Publishing isolation claims without multi-band rules or without COMFORT manoeuvre limits invites over-claiming and under-design of anti-roll behaviour.

REQ-M09/M10 exist because stops and power-loss behaviour are where magnetic systems typically fail buyer safety review. Numeric stroke/energy and an explicit S4 lock policy are therefore first-class, not appendix trivia.

REQ-M12/M23 exist because electromagnetic and contamination realities are easy to postpone and expensive to retrofit once hardware is frozen.

REQ-M13 exists because Path A without ICD is a change-order machine. REQ-M17 exists because false SPORT is a handling and liability hazard. REQ-M19 exists because three-path arbitration without priority becomes non-deterministic under concurrent inputs.

Annex AI — Risk Treatment Summary by Class

Risk class	Examples	Buyer-visible effect if open
Catastrophic	R-01, R-08	No safety credit; no integration sign-off
Handling/comfort high	R-03C, R-04S, R-02	No comfort % claims; limited demo scope
Supply/cost high	R-05, R-15	Commercial terms stay option-level
IP/export high	R-12, R-16, R-17	Disclosure throttled; counterparty filters
Medium closure	R-14, R-10	L2/G5 gates hold

Catastrophic items block any claim of readiness until analysis and design evidence exist. High items tied to COMFORT and S4 block comfort-led marketing until policies are numeric. IP and export items block broad disclosure and certain counterparties until counsel processes complete.

Annex AJ — Supplier RFQ Checklist (Path A)

- Minimum package to issue a responsible Path A RFQ after G1:
- Released ICD draft covering mechanical, hydraulic, electrical, communications, safety and lock domains.
 - Module datasheet with TARGET/STUDY labels intact (no silent conversion to guaranteed values).
 - State-machine description including S4 and lock confirm rules.
 - Environmental envelope (temperature, sealing intent, contamination).
 - Test outline for L2: force curve, power-loss landing, lock cycle, thermal RMS.
 - Quality and change-control expectations (even if CM system is still being stood up).
 - Export-control notice and NDA status of the recipient.

Annex AK — Test Outline Skeleton for L2 Single-Tower

Test	Objective	Pass idea (planning)
Force-gap / force-current	Map actuator authority	Curve covers ±2.5 kN study band or documents limit
Power-loss landing	S4 passive support	Settles within envelope; valves fail-closed
Lock engage/release	Confirm path	≤3 s class; positive confirm; no false SPORT
Thermal RMS	Continuous duty heating	Within ICD temperature limits
Contamination robustness	Seal/bellows integrity	No debris-induced jam in agreed trial
Stop contact	Energy path	Within STR-001 curve

Detailed procedures, instrumentation lists and uncertainty budgets are owned under test plans linked to L2 — not fully specified in Doc 00.

Annex AL — Four-Tower / HIL Outline for L3

L3 demonstrates multi-corner coherence: synchronized or intentionally asymmetric commands, mode transitions under load, power-loss from COMFORT and from SPORT, and stop engagement without loss of capture. HIL covers state-machine transitions with injected faults (loss of confirm, loss of supply, sensor freeze). Acceptance is structured, not anecdotal.

L3 theme	What is shown
Coherence	Four corners track within ICD limits
Mode transition	SPORT↔COMFORT ≤3 s under defined mass/road
Power-loss	From each mode into S4 policy
Fault injection	Confirm loss, supply loss, sensor freeze
Stops	Controlled engagement within design

Annex AM — Glossary Expanded

Term	Meaning
CMFSB	Captured Mechanical Floating Safety Body
Floating	Body isolation relative to chassis — not free levitation, not marine flotation
COMFORT	Soft isolation mode; locks released; full macro envelope
SPORT	Locked handling mode; upper body seated; stiffness rises
Reference path	Hybrid hydraulic + Lorentz
OPTION path	PM+EM requiring stability gate
L0–L4	Evidence ladder levels
Ph / G / Pri	Phase / Gate / Priority — exclusive naming
Line-sum P90	Sum of per-line P90 masses — conservative versus joint statistical P90
ICD	Interface Control Document
HIL	Hardware-in-the-loop
STR-001	Stop energy/stroke analysis work product
SIM-001	Multi-band isolation simulation work product

Annex AN — Distribution and Handling Rules

T1 distribution requires executed NDA. Printed excerpts shall carry the revision identifier ATFSV-DOC-00-R4.9. Recipients shall not strip footers. T2/T3 materials are need-to-know and counsel-controlled where claim charts are involved. Return or destruction of controlled copies follows the NDA terms.

Tier	Allowed content	Control
T0	Non-enabling teaser	Public
T1	This Doc 00	NDA
T2	Models, ICD detail, commercial terms	NDA + need-to-know
T3	Claim charts / IP-001	Counsel-controlled

Annex AO — Alignment of Gates, Phases and Evidence Levels

This annex aligns Phase (Ph), Gate (G) and Evidence level (L) so that programme language stays consistent (Finding 15).

Phase	Typical gate	Evidence level	Primary products
Ph0	G0–G1	L0	Envelope, ICD draft, vocabulary freeze
Ph1	G2–G3	L1–L2	SIM multi-band; single-tower bench
Ph2	G4	L3	Four-tower + HIL
Ph3	G5	L4	Optional mule metrics

Priority (Pri-1...3) orders work inside a phase; it is not a phase number and not a gate number. Using P0/P1/P2 for all three concepts caused confusion in earlier drafts and is forbidden in R4.9.

Annex AP — Cabin Acceleration Monitoring (optional instrumentation)

Optional cabin accelerometers support monitoring and post-event analysis. They are not required for the BASELINE power-loss path, which is hardware-dominant. If used for control, they enter the safety analysis and cybersecurity scope. If used only for monitoring, they remain outside the primary safety path but still require signal integrity appropriate to their role.

Annex AQ — Contamination Design Rules (Finding 11 expanded)

Gap and lock interfaces shall be protected by bellows or equivalent seals. Detection of degraded sealing or debris is TARGET. Environmental tests covering mud, ice and ferrous particle exposure are required before claiming field readiness. Inspection intervals shall be stated in service documentation. FMEA RPN 112 reflects the combination of severity and detection difficulty for gap contamination.

Interface	Protection	Detection	Test
Magnetic gap	Bellows/seals	Gap anomaly / visual	Mud/ice/ferrous trial
Lock faces	Sealed interface	Confirm degradation	Cycle + contaminant
Hydraulic rods	Wipers/seals	Leak / pressure	Environmental

Annex AR — Cybersecurity Touchpoints (planning)

If software can change force limits, mode logic, lock commands or update calibration, ISO/SAE 21434 and UN R155/R156 concepts become relevant for vehicle integration. Module-level mitigations include: signed software updates; safety partition for force and lock commands; rate limits; inhibit of high force when confirm is lost. Full vehicle cybersecurity case is Path B scope; module must still not present an open remote path to defeat capture or force unbounded commands.

Annex AS — Comparison Note: R4.8 vs R4.9 Density

R4.8 provided fourteen useful pages of module-primary baseline content. R4.9 preserves that mission, inserts corrections for Findings 1–23, retains structured tables, expands dual-mode, requirements rationale, risk treatment, mass method and SIM reporting, and presents a continuous dense layout. Empty pages are avoided by continuous flow; every page carries tables or substantive paragraphs.

Attribute	R4.8	R4.9
Architecture	Module-primary dual-mode	Same — retained
Numeric consistency	Issues listed in review	Corrected
SIM acceptance	Weak single floor	Multi-band + screening
S4 at speed	Under-specified	Lock + limp-home policy
COMFORT handling	Under-specified	Numeric policy required
Path B detail	Long in Doc 00	Summary; DOC-01
Commercial sensitivity	Country/price risk	Desensitised
Figures	Labels only	Drawn 1–4

Annex AT — Final Checklist Before External Release

- Confirm PDF metadata title is ATFSV-DOC-00-R4.9 CMFSB Module-Primary External Baseline.
- Confirm no USD prices or country-named counterparties appear in the technical body.
- Confirm Path B detailed structural content is summary-only.
- Confirm multi-band SIM rules and PRELIMINARY screening mark are present.
- Confirm S4 lock preference and limp-home policy are present.
- Confirm 3 g bands, PM bias gap rule, corner mass and fuel mass corrections are present.
- Confirm FMEA includes RPN and contamination rows.
- Confirm position statement is at the end and change log holds revision storytelling.
- Confirm NDA is executed before T1 distribution.

Closing Statement

ATFSV-DOC-00 Revision 4.9 is the corrected module-primary external baseline for technical review, NDA due diligence and supplier engagement. It preserves the useful substance of R4.8, corrects identified defects, and presents a dense continuous package without intentional blank pages.

Role	Organisation	Date	Signature
Author	ATFSV Engineering	2026-10-02	
Reviewer	TBD		
Approver	TBD		

END OF DOCUMENT — ATFSV-DOC-00-R4.9

Preliminary Engineering Baseline • Not Certified • TRL 2–3 • Confidential IP

R4.8 body preserved • Findings 1–23 corrected • Dense continuous layout • No intentional blank pages

Annex AU — Worked Energy Example for Isolation Power (planning)

Mean isolation power couples to the vehicle energy budget only when continuous duty is high. Screening PRELIMINARY force demand of 0.4–0.65 kN 3σ on road noise suggests that continuous RMS force well below 0.7 kN may suffice for straight-road isolation, while leveling pulses and manoeuvre loads dominate peaks. ATFSV-ENE-001 shall separate isolation mean power, leveling energy per event, and lock pulse energy.

Power element	Nature	Notes
Isolation mean	Continuous while COMFORT active	From SIM/test mean force × velocity stats
Leveling pulse	Event-driven	Four-corner settle energy
Lock pulse	Event-driven	Per engage/release
Electronics continuous	Continuous	Controllers/sensors
Pump average	Duty-cycle dependent	Leak + leveling frequency

Until these are measured, module electrical budget remains OPEN at L2. Do not advertise a single watt number as total module consumption.

Annex AV — Handling Default Policy Template (to be frozen)

The following template is a planning aid for REQ-M21 and is not a frozen calibration. Values in brackets are placeholders for the dynamics owner.

Policy element	Template	Owner
COMFORT max roll gradient	[X] °/g steady-state TARGET	Dynamics
Lateral-g inhibit to SPORT	Above [Y] g command lock/SPORT	Dynamics + Controls
Speed-dependent default	Above [Z] km/h default [SPORT/COMFORT]	Controls
Driver override	Always available except safety inhibit	Controls
False COMFORT inhibit	Yaw-rate / lat-g concurrent logic	Safety

Freezing this table is a G2 deliverable. Until frozen, COMFORT marketing and unrestricted COMFORT demos at high lateral demand are inappropriate.

Annex AW — Document History Integrity Statement

R4.9 was produced by retaining R4.8 module-primary structure and tables, applying Findings 1–23 corrections, adding drawn figure content, multi-band SIM rules, COMFORT and S4 policies, numeric crash requirements, expanded FMEA/risks/requirements traceability, desensitised commercial language, and Path B summary-only scope. The goal is a diligence-ready technical baseline that is denser and more consistent than R4.8 without discarding R4.8’s useful explanations.

This package intentionally exceeds fourteen pages because corrected and expanded material was added rather than substituted by compression. No page is intentionally left blank.

Annex AX — Quick Reference Card (printable summary)

Item	Value / rule
Architecture	Captured macro–micro; not free levitation
Modes	COMFORT + SPORT lock
Topology	Hydraulic + Lorentz BASELINE; PM+EM OPTION
Stroke / gap	±50 mm macro; ±7.5 mm micro; 15–30 mm gap
Force / mass	±2.5 kN study; ≤55 kg TARGET
3 g loads	S 9–15; M 15–21; L 21–24 kN
S4	Passive + prefer lock; limp-home ≤80 km/h class
SIM	Multi-band vs passive k,c; ISO 8608 B/C
GVW baseline	7,500 kg only
Path A	Module license/supply primary
Path B detail	ATFSV-DOC-01
TRL	2–3; not certified

Final Sign-Off Block

ATFSV-DOC-00-R4.9 is released for controlled distribution under NDA as the current module-primary external engineering baseline.

Role	Organisation	Date	Signature
Author	ATFSV Engineering	2026-10-02	
Reviewer	TBD		
Approver	TBD		

END OF DOCUMENT — ATFSV-DOC-00-R4.9

Preliminary Engineering Baseline • Not Certified • TRL 2–3 • Confidential IP

R4.8 body preserved • Findings 1–23 corrected • Dense continuous layout • Minimum 16 pages • No intentional blank pages

Annex AY — Record of Correction Authority (Findings 1–23 one-line)

#	Finding	R4.9 action
1	Figures missing	Drawn layer/state/power-loss figures added
2	SIM single floor	Multi-band + PRELIMINARY screening
3	COMFORT roll risk	Numeric policy + anti-roll requirements
4	S4 at speed	Lock preference + limp-home + HARA
5	Crash stop numeric	REQ-M09a/b/c
6	GVW mix	7,500 kg single baseline
7	Mass lines	Corner 560/640; fuel 40/55
8	Energy 816 vs 886	Reconciled; η 28–30%; fuel TARGET
9	3g inconsistency	S 9–15; M 15–21; L 21–24
10	Power incomplete	Full consumer table
11	FMEA no RPN	RPN + contamination + lock rows
12	PM bias no gap	Gap-dependent at 15 mm
13	REQ weak trace	Risk/ICD/gate columns
14	Risk gaps	Expanded register R-01...R-18
15	Naming collision	Ph / G / Pri / L exclusive
16	Vocabulary drift	Exclusive status table
17	Standards thin	Matrix expanded
18	Magnetic exposure	≤0.5 mT planning
19	Prior art modes	Counsel caution; differentiate lock+ICD
20	Path B overload	Summary only; DOC-01
21	Commercial sensitivity	No countries/prices in Doc 00
22	Structure	Position at end; change log
23	PDF quality	Metadata; dense continuous layout

Each action is reflected in the main body or annexes. Appendix C remains the diligence checklist. This annex is the one-line executive map.

Document control: ATFSV-DOC-00-R4.9 supersedes R4.8. Controlled distribution under NDA. Not certified. TRL 2–3.

Role	Organisation	Date	Signature
Author	ATFSV Engineering	2026-10-02	
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16 pages • R4.8 substance preserved • Findings 1–23 corrected • No intentional blank pages